

GSM8K · \$5.1

Qwen2.5-14B-Instruct · lm-evaluation-harness 0.4.11 · 5-shot

Plain few-shot (direct-answer)

[sys] You are Qwen, created by Alibaba Cloud.

Q: Jen and Tyler are gymnasts ...

A: ... #### 12

Q: Four people in a law firm ...

A: ... #### 1

... [3 more few-shot examples] ...

Q: Janet's ducks lay 16 eggs/day.

She eats 3 for breakfast; bakes 4 into muffins.

She sells the rest at \$2 each. How much?

A: 16-3-4 = 9 eggs. 9 × 2=18. #### 18

58.5%

↓ Δ = -56.0 pp

Chain-of-thought few-shot

[sys] You are an expert problem solver.

Think step by step, then state the answer.

Q: Jen and Tyler are gymnasts ...

A: ... #### 12

... [same 5 few-shot examples] ...

Q: Janet's ducks lay 16 eggs/day. ...

A: Janet uses 3 + 4 = 7 eggs each day.

16 - 7 = 9 eggs remain.

9 × 2=18 per day.

The answer is \$18 per day.

2.5%

Same model · same harness · same 5-shot context.
CoT prompt elicits step-by-step prose;
the #### delimiter regex finds nothing.
Parser records 0 for correctly-answered items.

Prompts: run logs · experiments/controlled_eval/5-shot-GSM8K/
Scores verified: 58.5% plain, 2.5% CoT (limit=200, seed=42/123/7)

Belebele · \$5.2

Mistral-7B · two papers · two evaluation regimes

Reka paper · 0-shot · multilingual avg (150 langs)

0-shot multilingual template (Reka)

(canonical; verbatim not published)

Passage: [reading comprehension context]

Question: What does the passage say about ...?

A) [option] B) [option]

C) [option] D) [option]

Answer:

← 0-shot; no few-shot examples in context

← average over 150 language variants

Mistral-7B score: 32.80

32.80

↑ Δ = +46.6 pp

Falcon2 report · 5-shot · English only

5-shot English-only template (Falcon2)

(canonical; verbatim not published)

Example 1: [English passage + question]

Answer: B

... [4 more few-shot examples] ...

Passage: [English reading comprehension]

Question: What does the passage say about ...?

A) [option] B) [option]

Answer: ← 5-shot; English only

Mistral-7B score: 79.42

79.42

Same benchmark label · two dimensions differ.
Language scope: multilingual avg vs. English only.
N-shot: 0-shot vs. 5-shot.
Falcon2 internal 0-shot English = 32.48 ≈ Reka 32.80;
n-shot, not language scope, drives the 46.6 pp gap.

Scores: arXiv:2404.12387 (Reka, 32.80)
arXiv:2407.14885 (Falcon2, 79.42 · internal 0-shot = 32.48)

BBH · Gemma-2-9B · \$5.3

Five sources · two score clusters · two protocols

OLv2 & SEA-LION · lighteval · 3-shot · log-likelihood

lighteval BBH protocol (OLv2 / SEA-LION)

Q: Which statement is logically false?

given: all mammals breathe air;

whales are mammals.

(A) Whales breathe air. (B) Whales swim.

(C) Whales are reptiles. (D) Whales have lungs.

rank: argmax log P(choice | context)

← 3-shot; log-likelihood ranking

OLv2 = 34.10 ↱ exact 2-decimal match

SEA-LION = 34.10 ↲ → shared pipeline origin

34.10

↑ Δ = +34.6 pp

OLMoE / SmoLLM2 / RecurrentGemma · 0-shot CoT

0-shot CoT protocol (Suzgun et al. 2023)

Q: Which statement is logically false? ...

Let's think step by step.

All mammals breathe air; whales are mammals

therefore whales breathe air. That is true.

Whales are not reptiles – that is false.

The answer is (C).

OLMoE = 68.20 SmoLLM2 = 69.00

RecurrentGemma = 69.00 (independent)

68.73

Two score clusters — two protocols.
OLv2 & SEA-LION: exact 2-decimal match;
shared pipeline, not independent evidence.
OLMoE/SmoLLM2/RecurrentGemma: 0-shot CoT.

Scores: analysis_output/validation_pairs.txt
arXiv:2504.05747 (SEA-LION, Table 3)
BBH cluster 2 mean: (68.20+69.00+69.00)/3 = 68.73